

Lecture 9: Linear discriminant analysis

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116 Advanced Statistics, SS 2025

Credits:

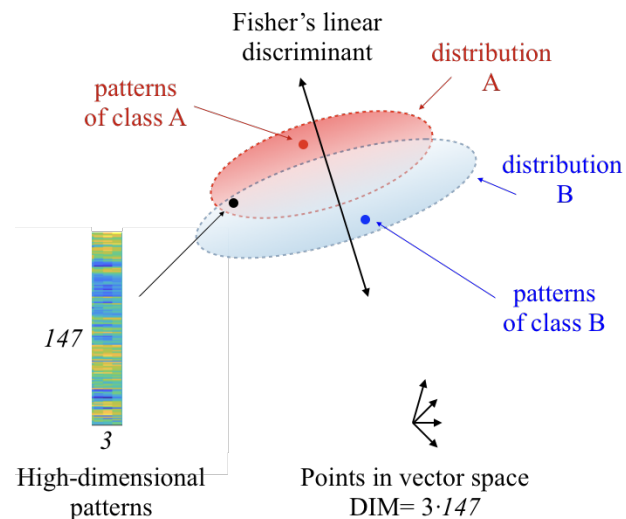
Chapter 12 in “Statistical Learning and Machine Learning with R”,
teazrq.github.io/SMLR/

Anderson (2001) “A new method for non-parametric multivariate
analysis of variance.” *Austral Ecology*

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1. Linear discriminant analysis



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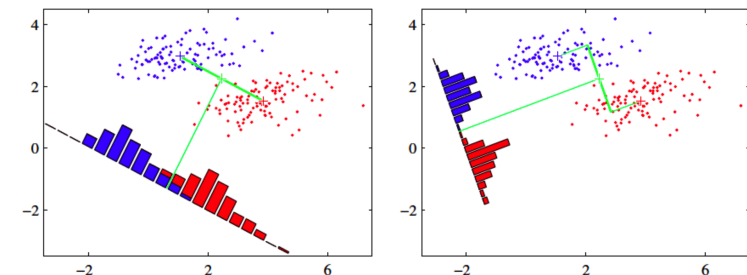
Lecture 9

1. Linear discriminant analysis
2. Example: perceptual development of females and males.
3. Multidimensional analysis of variance
4. Example: adolescent EEG states
5. Diligent data inspection

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Most discriminating projection axis (2D example)



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Centroids and covariance matrices

Given two classes of vector observations $\mathbf{x} \in X_1$ and $\mathbf{x} \in X_2$ in a vector space $\mathcal{R}^n$, we can obtain the centroid for each class (mean vectors)

$$\mathbf{m}_1 = \frac{1}{n_1} \sum_{\mathbf{x} \in X_1} \mathbf{x}, \quad \mathbf{m}_2 = \frac{1}{n_2} \sum_{\mathbf{x} \in X_2} \mathbf{x}$$

and the covariance matrix for each class

$$\mathbf{C}_1 = \frac{1}{n_1} \sum_{\mathbf{x} \in X_1} (\mathbf{x} - \mathbf{m}_1)(\mathbf{x} - \mathbf{m}_1)^T$$

$$\mathbf{C}_2 = \frac{1}{n_2} \sum_{\mathbf{x} \in X_2} (\mathbf{x} - \mathbf{m}_2)(\mathbf{x} - \mathbf{m}_2)^T$$

Maximally discriminating axis

We wish to find the projection $\mathbf{w}$ that maximizes the objective function

$$\frac{m_1 - m_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

which quantifies the discriminability of the two sets of projected values in terms of a d' or z-score value.

Projection onto axis

Given any projection axis $\mathbf{w}$ with unit norm, $\|\mathbf{w}\| = 1$, we can use the dot product to compute the scalar projection $x \in \mathcal{R}$ of vectors $\mathbf{x}$ onto axis $\mathbf{w}$:

$$x = \mathbf{x} \cdot \mathbf{w} = \mathbf{x}^T \mathbf{w}$$

The class means of the projected values are

$$m_1 = \mathbf{m}_1^T \mathbf{w}, \quad m_2 = \mathbf{m}_2^T \mathbf{w}$$

and the class variances of the projected values are

$$\sigma_1^2 = \mathbf{w}^T \mathbf{C}_1 \mathbf{w}, \quad \sigma_2^2 = \mathbf{w}^T \mathbf{C}_2 \mathbf{w}$$

Fisher's linear discriminant

To find projection $\mathbf{w}$, we form summed within-class variance with

$$\mathbf{C}_W = \mathbf{C}_1 + \mathbf{C}_2$$

obtain inverse matrix $\mathbf{C}_W^{-1}$ (which in some cases may be numerically tricky) and multiply with the mean between-class difference vector

$$\mathbf{y} = \mathbf{C}_W^{-1} (\mathbf{m}_1 - \mathbf{m}_2) = (\mathbf{m}_1 - \mathbf{m}_2)^T \mathbf{C}_W^{-1}$$

and normalize the result to unity

$$\mathbf{w} = \frac{\mathbf{y}}{\|\mathbf{y}\|}$$

This most discriminating direction is called "Fisher's linear discriminant".

Finally, to assess the quality of this 'optimal discriminant', we project (as described above) onto Fisher's discriminant $\mathbf{w}$ and obtain the class means

$$m_1 = \mathbf{m}_1^T \mathbf{w}, \quad m_2 = \mathbf{m}_2^T \mathbf{w}$$

and variances

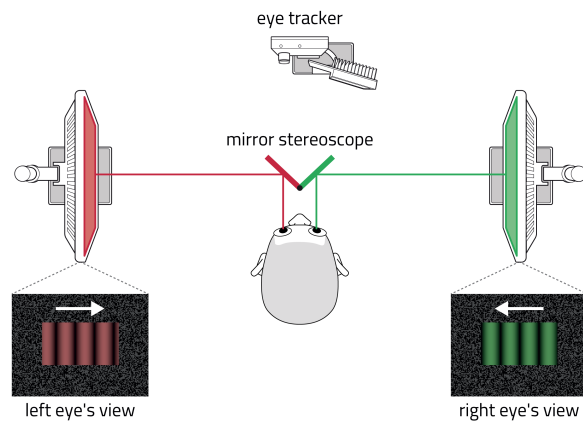
$$\sigma_1^2 = \mathbf{w}^T \mathbf{C}_1 \mathbf{w}, \quad \sigma_2^2 = \mathbf{w}^T \mathbf{C}_2 \mathbf{w}$$

and use the z-score

$$z_{norm} = \frac{m_1 - m_2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

as a parametric test for significance. If we prefer a non-parametric test, we can instead use a Wilcoxon (Mann-Whitney) test and Z_{wlcxn} .

2. Example: Perceptual development of females and males



Ziman, Aleshin, Unoka, Braun, Kovacs (2022) Alternative female and male strategies in the dynamic balance of visual perception. *Scientific Reports*, **12**: 1674.

Crossvalidation

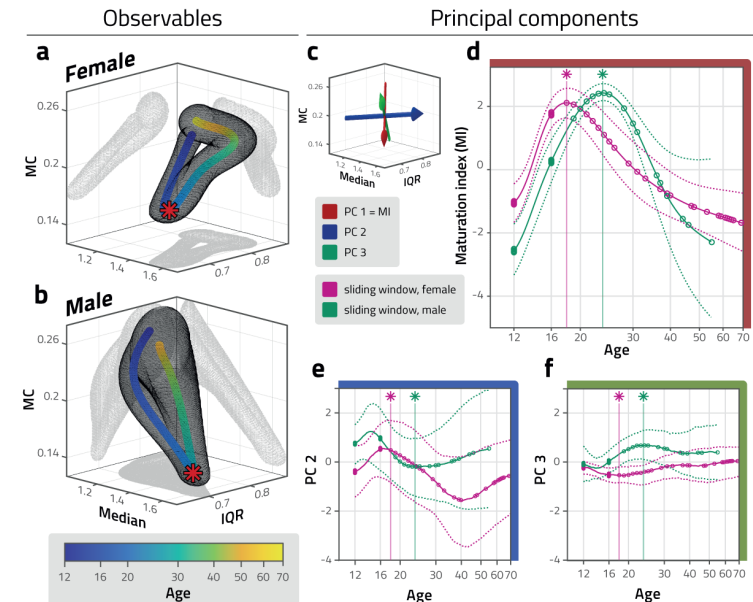
Kriegeskorte & Diedrichsen (2019, Annu Rev Neurosci) crossvalidate this approach by dividing events into training and test sets. They call

$$c^2 = \mathbf{w} (\mathbf{m}_1^{test} - \mathbf{m}_2^{test}) = (\mathbf{m}_1 - \mathbf{m}_2)^T \mathbf{C}_W^{-1} (\mathbf{m}_1^{test} - \mathbf{m}_2^{test})$$

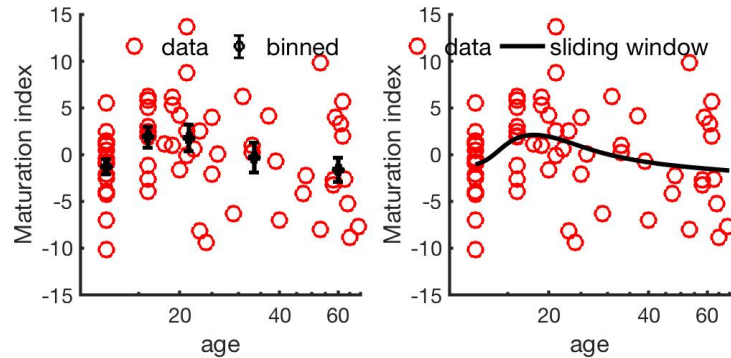
the 'crossnobis estimator', as it is a crossvalidated version of the Mahalanobis distance

$$d^2 = (\mathbf{m}_1 - \mathbf{m}_2)^T \mathbf{C}_W^{-1} (\mathbf{m}_1 - \mathbf{m}_2)$$

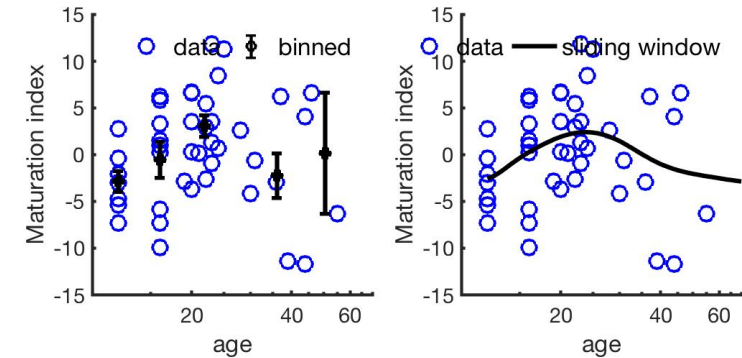
Alternative developmental trajectories of binocular rivalry



Female observations, with discrete and continuous average



Male observations, with discrete and continuous average

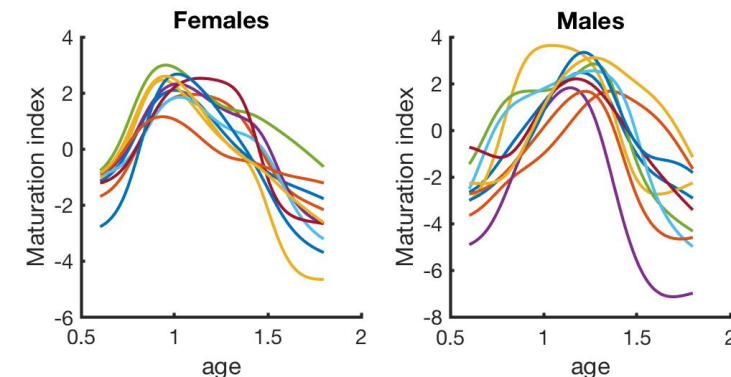


Are developmental trajectories of females and males significantly different?

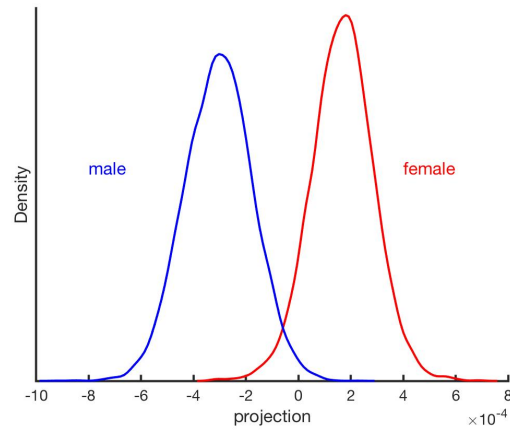
- ▶ Bootstrap I: Sample female **or** male observers randomly (with replacement) to obtain many female **or** male trajectories.
- ▶ Linear discriminant analysis I: represent trajectories in 20-dim vector space and obtain distribution of pairwise distances.
- ▶ Bootstrap II: Sample female **and** male observers randomly (with replacement) to obtain many pseudo-female or pseudo-male trajectories.
- ▶ Linear discriminant analysis II: represent trajectories in 20-dim vector space and obtain distribution of pairwise distances.
- ▶ Compare distributions for real-sexes and for pseudo-sexes with parametric z-score or non-parametric signed Wilcoxon rank-sum test.

Examples of bootstrapped 'true-sex' trajectories

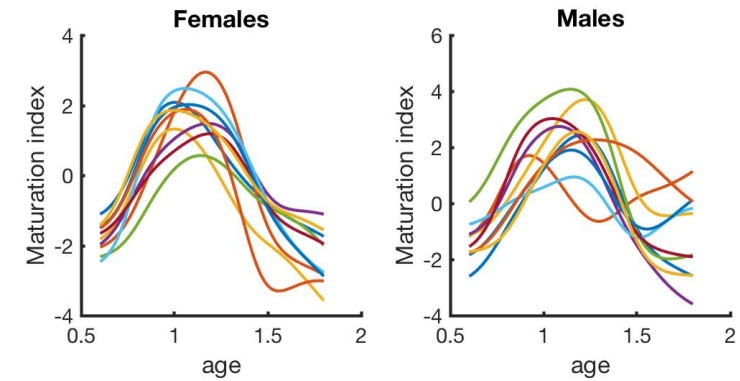
Every trajectory becomes a point in 20-dimensional space. Male and female trajectories form two 'clouds' in this space.



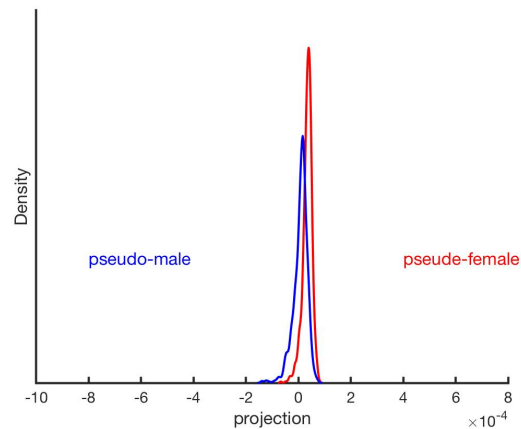
True-sex trajectories projected onto most discriminant axis



Examples of bootstrapped 'pseudo-sex' trajectories

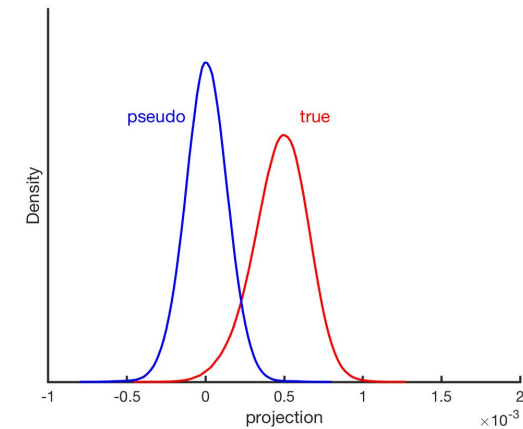


Pseudo-sex trajectories projected onto most discriminant axis



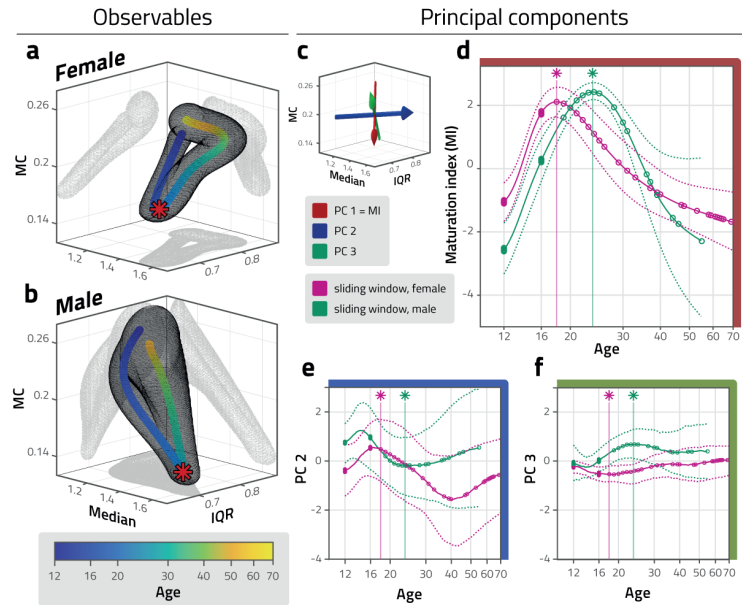
Comparison of pairwise distances

Pairwise distances for true-sex and pseudo-sex trajectories.



Parametric z-score $z = 2.11$ and p-value $p < 0.0174$. We conclude that the development of average females differs significantly from that of average males.

Alternative developmental trajectories of binocular rivalry



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Within-class variance

Consider points $x_{ik} \in \mathcal{R}^n$ where k indexes a class of points and i indexes its members. The class centroids and overall centroids are

$$c_k \equiv \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ik}, \quad c_{\text{overall}} \equiv \sum_{k=1}^K \frac{n_k}{N} c_k, \quad N = \sum_{k=1}^K n_k$$

The “within-class” sum of squares

$$SS_W^{\text{cent}} = \sum_{k=1}^K SS_{Wk}^{\text{cent}}, \quad SS_{Wk}^{\text{cent}} = \sum_{i=1}^{n_k} (x_{ik} - c_k)^2$$

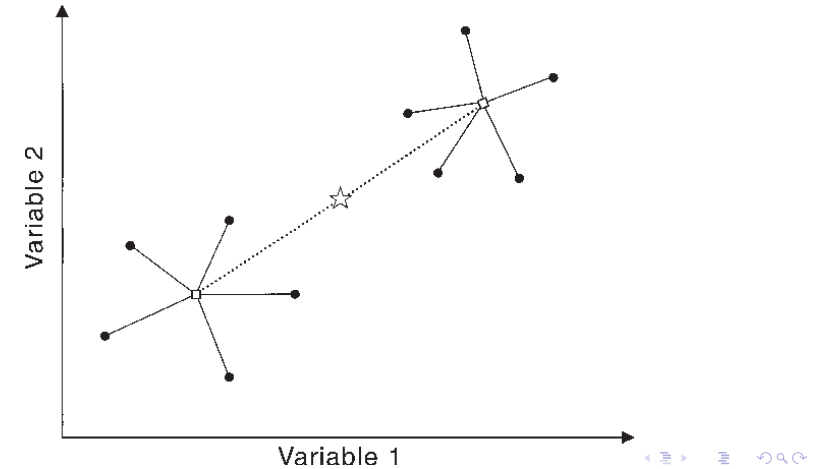
equals sum of squared distances between class centroid c_k and class members k , summed over all K classes. The total number of distances is $N = \sum_{k=1}^K n_k$.

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3. Multidimensional analysis of variance (Anderson, 2001)

Consider multiple classes of exemplars $x \in \mathcal{R}^n$, their class centroids, and the overall centroid. How much variance does this particular classification explain?



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Between-class variance

The “between-class” sum of squares

$$SS_B^{\text{cent}} = \sum_{k=1}^K SS_{Bk}^{\text{cent}}$$

$$SS_{Bk}^{\text{cent}} = \sum_{i=1}^{n_k} (c_k - c_{\text{overall}})^2 = n_k (c_k - c_{\text{overall}})^2$$

is the sum of squared distances between class centroid k and global centroid. Note that we are still summing over all points x_{ik} , except that *every point* x_{ik} is replaced by its class centroid c_k , so that the total number of distances is again $N = \sum_{k=1}^K n_k$.

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Total variance

The “total” sum of squares

$$SS_T^{cent} = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_{overall})^2$$

is the sum of squared distances between individual points and the overall centroid. The total number of distance is again

$$N = \sum_{k=1}^K n_k$$

Proof of variance summation:

Basically, squared distances between points and class centroids add to squared distances between class centroid and overall centroid:

$$\begin{aligned} \sum_{i=1}^{n_k} (x_{ik} - c_{overall})^2 &= \sum_{i=1}^{n_k} (x_{ik} - c_k + c_k - c_{overall})^2 = \\ &= \sum_{i=1}^{n_k} (x_{ik} - c_k)^2 + n_k (c_k - c_{overall})^2 + 2(c_k - c_{overall}) \underbrace{\sum_{i=1}^{n_k} (x_{ik} - c_k)}_{=0} \\ \sum_{i=1}^N (x_i - c_{overall})^2 &= \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_{overall})^2 = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_k)^2 + \sum_{k=1}^K n_k (c_k - c_{overall})^2 \end{aligned}$$

Dividing variance

A bit of reflection makes clear that

$$SS_T^{cent} = SS_W^{cent} + SS_B^{cent}$$

provided the number of squared distances in all sums is $N = \sum_{k=1}^K n_k$. Accordingly, we have

$$SS_B^{cent} = SS_T^{cent} - SS_W^{cent}, \quad \forall n_k$$

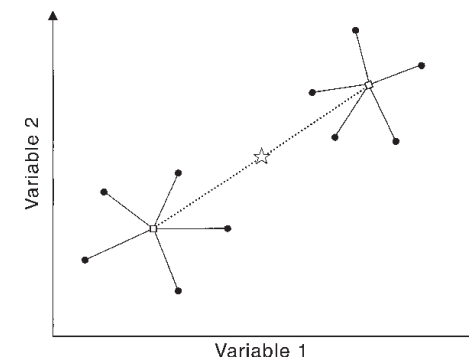
or alternatively

$$SS_B^{cent} = \sum_{k=1}^K n_k (c_k - c_{overall})^2 = \sum_{k=1}^K \sum_{l=k}^K \frac{n_k n_l}{N} (c_k - c_l)^2$$

A proof for the last equivalence is supplied in the next section.

Variance explained by classification

The the point-to-centroid approach motivates the notion of “explained variance”. How much variance is “explained” by the deviation of centroids from each other or, equivalently, from the overall centroid (“between-class”)? How much variance is “unexplained” and due the deviation of points from class centroids (“within-class”).



Anderson's F-ratio

Anderson's F-ratio is defined as the ration of "explained" to "unexplained" variance

$$F = \frac{SS_B^{cent} / (K - 1)}{SS_W^{cent} / (N - K)}$$

It quantifies the discriminability between classes, compared to the discriminability of exemplars within classes. Values near unity indicate indiscriminable classes, values above unity indicate highly discriminable classes.

SS_B is normalized to $K - 1$, as there are K centroid-to-centroid distances, and thus $K - 1$ degrees of freedom. SS_W is normalized to $N - K$, as there are K times n point-to-centroid distances, each with $n - 1$ degrees of freedom. Correspondingly, SS_T should be normalized to $N - 1$.

Navigation icons

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Sum of squared point-to-centroid distances and sum of squared point-to-point distances $d_{ij}^2 = (x_i - x_j)^2$:

$$\sum_{i=1}^{n_x} (x_i - c_x)^2 = \frac{1}{n_x} \sum_{i=1}^{n_x} \sum_{j=1}^{n_x} (x_i - x_j)^2, \quad c_x = \frac{1}{n_x} \sum_j x_j$$

when we sum over all unordered pairs ($i \leq j \leq n_x$), or

$$\sum_{i=1}^{n_x} (x_i - c_x)^2 = \frac{1}{2n_x} \sum_{i=1}^{n_x} \sum_{j=1}^{n_x} (x_i - x_j)^2$$

when we sum over all ordered pairs ($1 \leq j \leq n_x$).

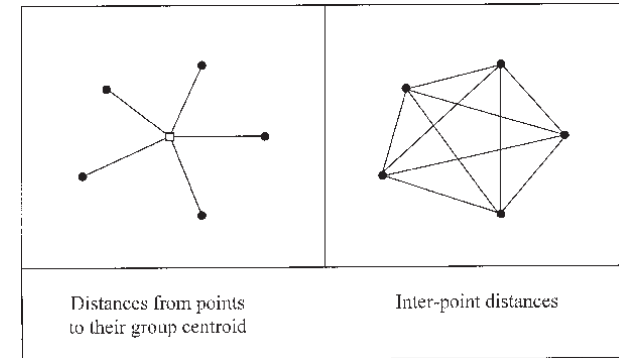
Navigation icons

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Pairwise point-to-point distances (within)

For an alternative geometrical intuition, note that the sum of squared point-to-centroid distances is equal to the sum of squared point-to-point distances, divided by the number of points.



Navigation icons

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Proof:

$$\begin{aligned} \sum_i \sum_j (x_i - x_j)^2 &= \sum_i \sum_j (x_i^2 - 2x_i x_j + x_j^2) = \sum_i \left[nx_i^2 - 2x_i \sum_j x_j + \sum_j x_j^2 \right] = \\ &= n \sum_i x_i^2 - 2 \left(\sum_i x_i \right)^2 + n \sum_i x_i^2 = \\ &= 2n \sum_i x_i^2 - 2 \left(\sum_i x_i \right)^2 = \\ &= 2n \left[\sum_i x_i^2 - \frac{2}{n} \left(\sum_i x_i \right)^2 + \frac{1}{n} \left(\sum_i x_i \right)^2 \right] = \\ &= 2n \sum_i \left[x_i^2 - \frac{2}{n} x_i \left(\sum_i x_i \right) + \frac{1}{n^2} \left(\sum_i x_i \right)^2 \right] = \\ &= 2n \sum_i \left(x_i - \frac{1}{n} \sum_j x_j \right)^2 \end{aligned}$$

Navigation icons

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Pairwise point-to-point distances (between)

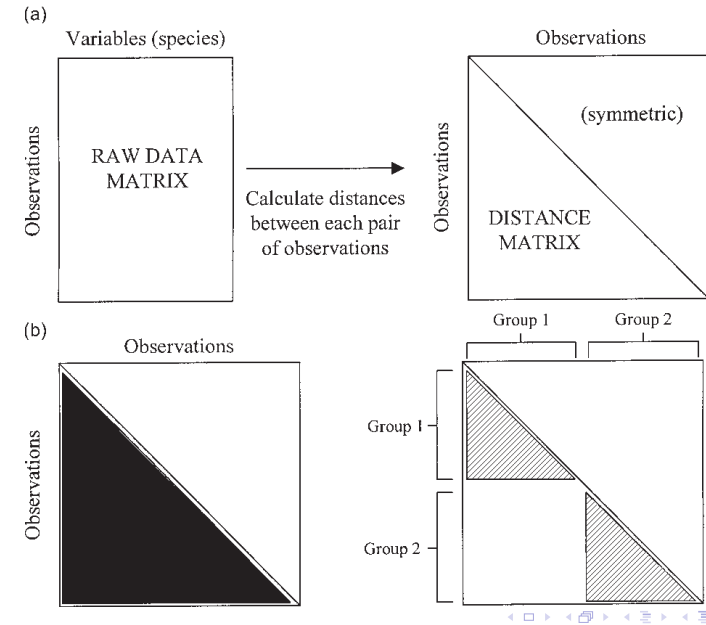
For two non-overlapping sets of points x_i and y_j with centroids

$$c_x = \frac{1}{n_x} \sum_{i=1}^{n_x} x_i, \quad c_y = \frac{1}{n_y} \sum_{j=1}^{n_y} y_j$$

we find

$$\begin{aligned} \sum_i (x_i - y)^2 &= \sum_i (x_i - c_x)^2 + n_x (y - c_x)^2 \\ \sum_j \sum_i (x_i - y_j)^2 &= \sum_j \left[\sum_i (x_i - c_x)^2 + n_x (y_j - c_x)^2 \right] = \\ &= n_y \sum_i (x_i - c_x)^2 + n_x \sum_j (y_j - c_x)^2 = \\ &= n_y \sum_i (x_i - c_x)^2 + n_x \sum_j (y_j - c_y)^2 + n_x n_y (c_x - c_y)^2 \end{aligned}$$

Dividing sums-of-squares of point-to-point distances



Dividing sums-of-squares of point-to-point distances

By construction

$$\begin{aligned} \underbrace{\frac{1}{N} \sum_{k=1}^N \sum_{k'=k}^N (z_k - z_{k'})^2}_{SS_{Tz}^{pair}} &= \underbrace{\frac{1}{N} \sum_{i=1}^{n_x} \sum_{i'=i}^{n_x} (x_i - x_{i'})^2}_{SS_{Wx}^{pair}} + \underbrace{\frac{1}{N} \sum_{j=1}^{n_y} \sum_{k'=k}^{n_y} (y_j - y_{j'})^2}_{SS_{Wy}^{pair}} + \\ &+ \underbrace{\frac{1}{N} \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} (x_i - y_j)^2}_{SS_{Bxy}^{pair}} \end{aligned}$$

Note that “total” and “within” sums are over unique pairs, whereas “between” sums are over all pairs (i.e., we are summing over all pairs below the diagonal).

Using the equivalences from the previous section, this can be related to the sums of squares of point-to-centroid distances

$$\begin{aligned} \underbrace{\sum_{k=1}^N (z_k - c_z)^2}_{SS_{Tz}^{pair}} &= \underbrace{\frac{n_x}{N} \sum_{i=1}^{n_x} (x_i - c_x)^2}_{SS_{Wx}^{pair}} + \underbrace{\frac{n_y}{N} \sum_{i=1}^{n_y} (y_i - c_y)^2}_{SS_{Wy}^{pair}} \\ &+ \underbrace{\frac{n_y}{N} \sum_i (x_i - c_x)^2 + \frac{n_x}{N} \sum_j (y_j - c_y)^2 + \frac{n_x n_y}{N} (c_x - c_y)^2}_{SS_{Bxy}^{pair}} \end{aligned}$$

$$\underbrace{\sum_{k=1}^N (z_k - c_z)^2}_{SS_T^{cent}} = \underbrace{\sum_{i=1}^{n_x} (x_i - c_x)^2}_{SS_{Wx}^{cent}} + \underbrace{\sum_{i=1}^{n_y} (y_i - c_y)^2}_{SS_{Wy}^{cent}} + \underbrace{\frac{n_x n_y}{n_x + n_y} (c_x - c_y)^2}_{SS_B^{cent}}$$

Generalization to K classes

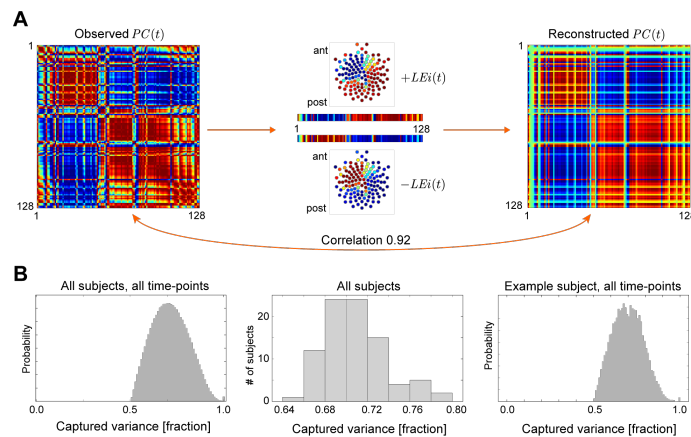
$$SS_T^{cent} = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_{overall})^2 = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N (x_i - x_j)^2 = SS_T^{pair}, \quad N = \sum_{k=1}^K n_k$$

$$SS_{Wk}^{cent} = \sum_{i=1}^{n_k} (x_{ik} - c_k)^2 = \frac{1}{n_k} \sum_{i=1}^{n_k} \sum_{j=1}^{n_k} (x_{ik} - x_{jk})^2 = \frac{N}{n_k} \left[\frac{1}{N} \sum_{i=1}^{n_k} \sum_{j=1}^{n_k} (x_{ik} - x_{jk})^2 \right] = \frac{N}{n_k} SS_{Wk}^{pair}$$

$$SS_W^{cent} = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_k)^2 = \sum_{k=1}^K \frac{1}{n_k} \sum_{i=1}^{n_k} \sum_{j=1}^{n_k} (x_{ik} - x_{jk})^2 = \sum_{k=1}^K \frac{N}{n_k} SS_{Wk}^{pair}$$

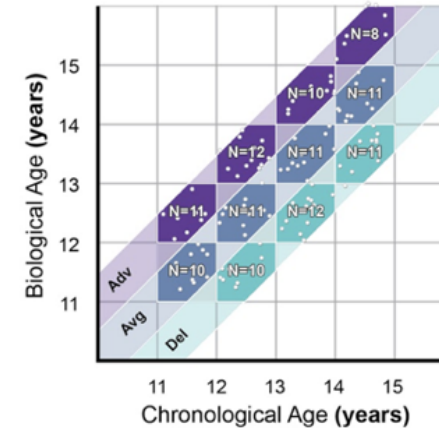
$$SS_B^{cent} = \sum_{k=1}^K n_k (c_k - c_{overall})^2 = \sum_{k=1}^K \sum_{l=1}^K \frac{n_k n_l}{N} (c_k - c_l)^2$$

Summarizing brain state with Leading Eigenvector (LEIDA)



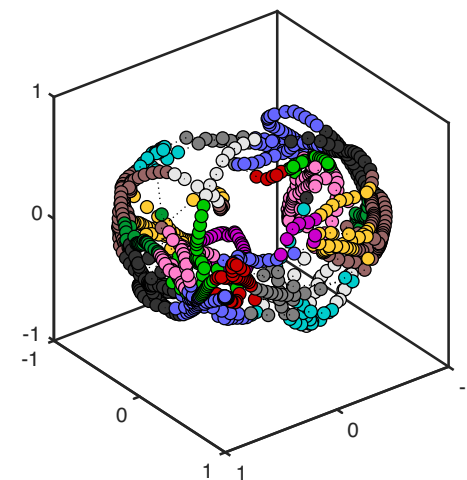
Principle of Leading Eigenvector Analysis (LEIDA). **A)** Representative example of observed phase coherence matrix $PC(t)$ and matrix reconstructed from leading eigenvector (matrix correlation $\rho = 0.92$, fraction of variance captured 70%). Color scale represents coherence values $-1 \leq \cos \Delta\Phi \leq +1$. The leading eigenvector $\pm LEi(t)$ is depicted both as linear arrays and as scalp distributions (with frontal pole towards the top). **B)** Distribution of captured variance fraction across the whole population of 87 subjects and all time-points. **C)** Distribution of captured variance, over all subjects and time-points (left), over individual subjects (middle), and over time-points of a representative individual subject (mean 0.70, standard deviation 0.09, range 0.50 to 1.0).

4. Example: adolescent EEG brain states



117 adolescent girls, categorized by chronological and biological age, 128 channel EGI EEG, 5 min eyes open and 5 min closed, processed into phase angles sampled at 100 Hz (30,000 intervals of 10 ms per subject).

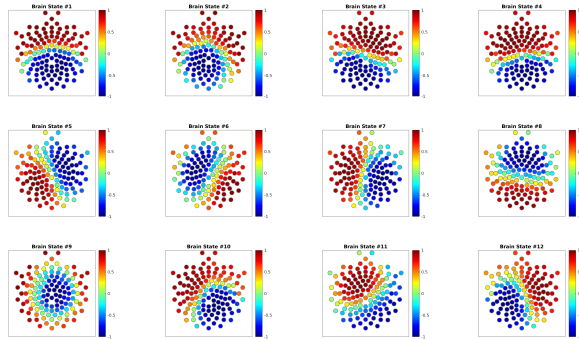
Sequence of 'brain states'



At every time point (10 ms), we obtain a 128×128 matrix of phase angle differences. The leading eigenvector of this matrix ("LEIDA") provides a concise summary of this matrix and reflects on the instantaneous 'brain state'. Here is an example sequence of such LEIDAs (projected onto the layout of EEG electrodes)

Classification of 'brain states'

Combining observations from all observers, we can use K-means to group 'brain states' into distinct classes.



Shown here are class prototypes (K-means centroids) for $N = 12$ classes.

Anderson's F-ratio

Between-class sum of squares

$$SS_B^{cent} = \sum_{k=1}^K n_k (c_k - c_{overall})^2$$

Within-class sum of squares

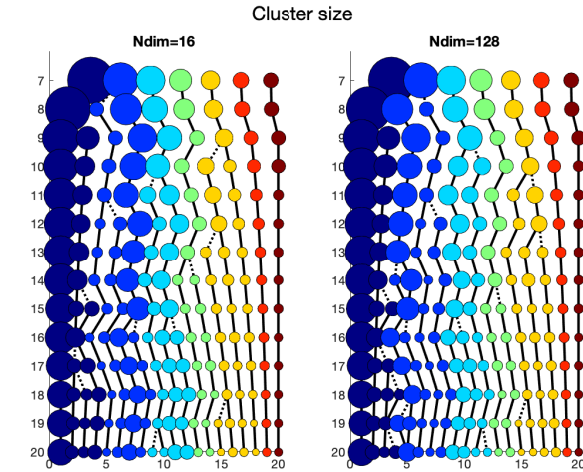
$$SS_W^{cent} = \sum_{k=1}^K \sum_{i=1}^{n_k} (x_{ik} - c_k)^2$$

F-ratio

$$F = \frac{SS_B^{cent} / (K - 1)}{SS_W^{cent} / (N - K)}$$

$$c_k \equiv \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ik}, \quad c_{overall} \equiv \sum_{k=1}^K \frac{n_k}{N} c_k, \quad N = \sum_{k=1}^K n_k$$

Which number of clusters should we choose?



Proximity of prototypes at different levels of clusterization (7 to 20). Color indicates a 'family', defined by proximity. Size indicates number of exemplars.

Silhouette coefficient

The silhouette value for each point is a measure of how similar that point is to points in its own cluster, when compared to points in other clusters. The silhouette value for the i th point, S_i , is defined as

$$S_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad a_i = \frac{1}{n_k - 1} \sum_{j=1}^{n_k} \|x_{ik} - x_{jk}\|^2$$

$$b_i = \min_{l \neq k} \frac{1}{n_l} \sum_{j=1}^{n_l} \|x_{ik} - x_{jl}\|^2$$

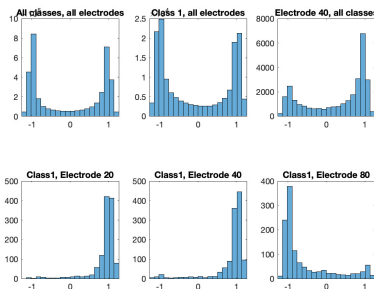
where a_i is the average distance from the i th point to the other points in the same cluster as i , and b_i is the minimum average distance from the i th point to points in a different cluster, minimized over clusters.

Silhouette ctd

The silhouette value ranges from -1 to +1. A high silhouette value indicates that x_{ik} is well-matched to its own cluster k , and poorly-matched to neighboring clusters $l \neq k$. If most points have a high silhouette value, then the clustering solution is appropriate. If many points have a low or negative silhouette value, then the clustering solution may have either too many or too few clusters. The silhouette clustering evaluation criterion can be used with any distance metric.

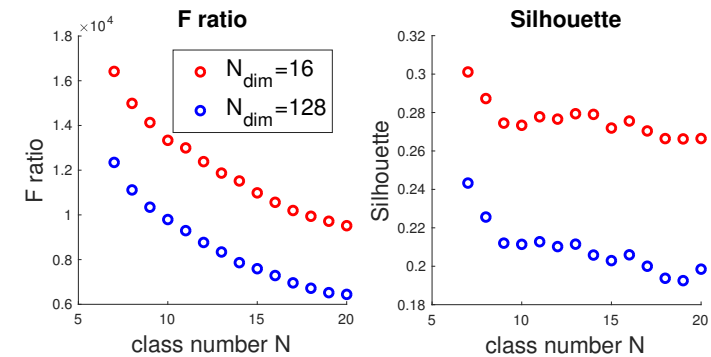
5. Diligent data inspection

It is not advisable to place blind trust into any computational procedure (such as eigenvalue analysis or K-means classification). Instead, it often helps to visualize your data in as many different ways as possible. A custom-made visualization increases confidence in computational results.



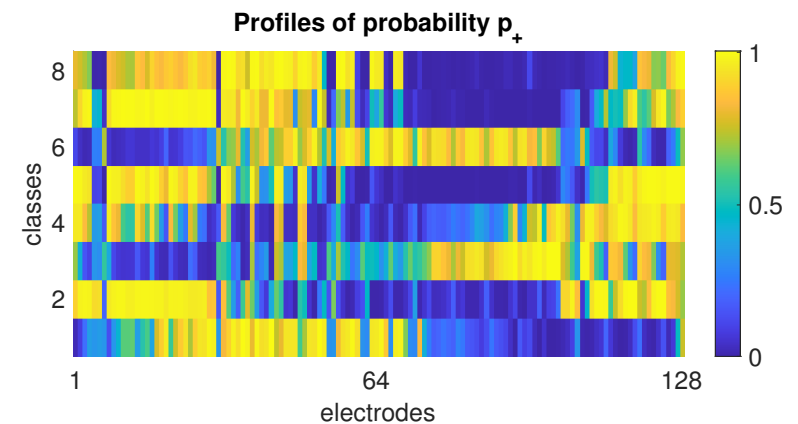
The element values of phase angles / LEIDA vectors cluster around positive and negative unity.

Results

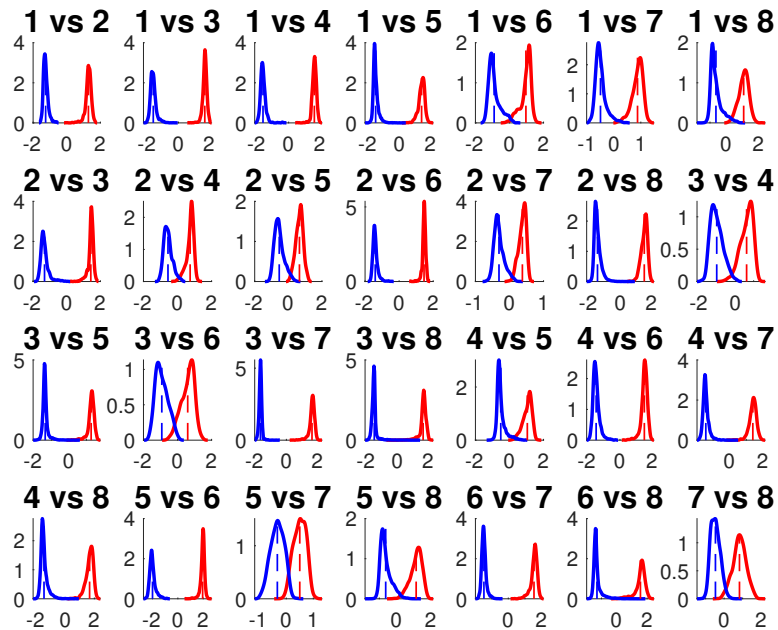


Reduced dimensionality improves quality of classification. F-ratio indicates very high quality. Silhouette indicates a slight advantage around $N = 12$.

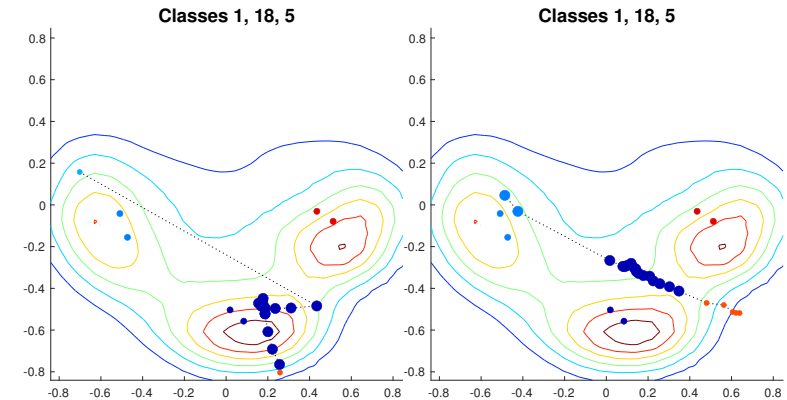
Nearly binary patterns of positive and negative angles



All electrodes, $N = 8$ classes. Next page: LDA shows high discriminability for all pairs of classes.

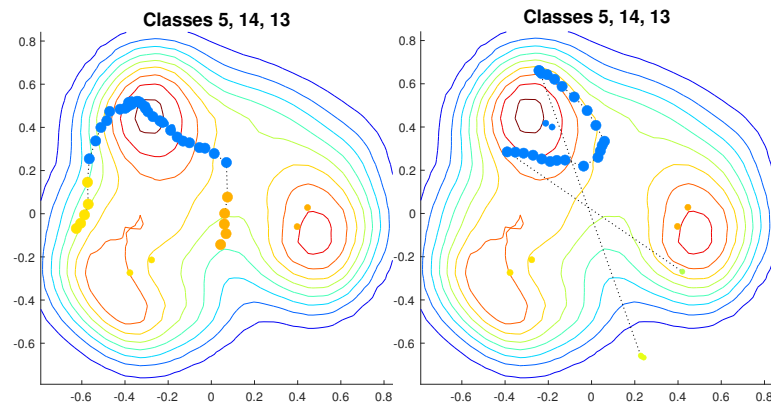


Trajectory examples



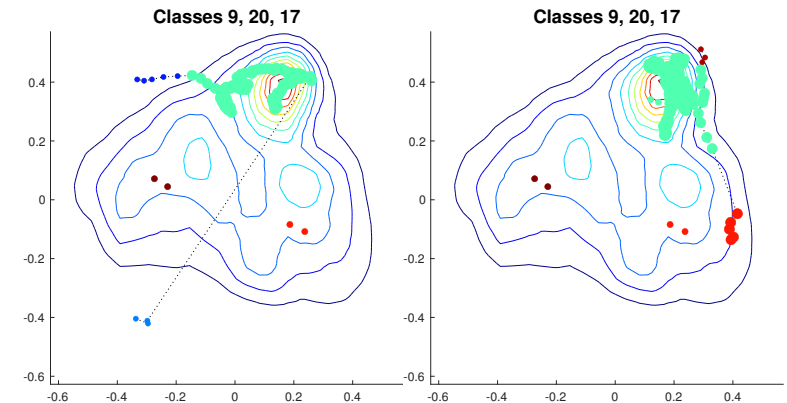
Two example trajectories, dwelling in central class (large dots, 10 *ms* steps) and transitioning to or from other classes (dotted lines, small dots). Densities of three classes are shown, each with two observer centroids (small dots). Projection into maximally discriminating plane for three classes.

Trajectory examples



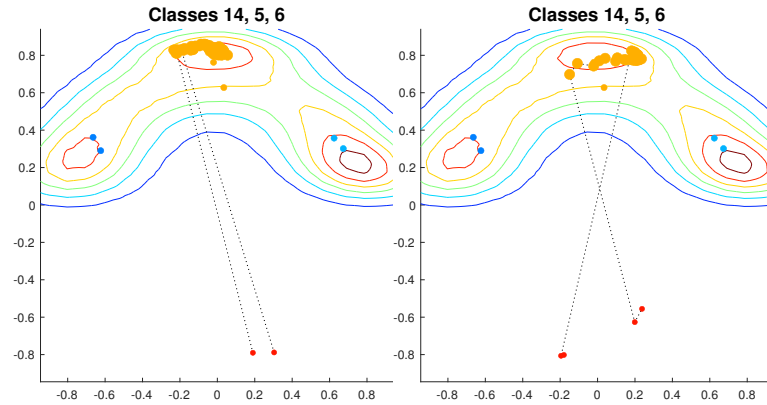
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The End

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